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Identification of Anxiety and Depression Using DASS-21 Questionnaire and Machine Learning

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***Abstract*— Identification of psychologically associated emotional activities through machine learning techniques and artificial intelligence is found to be widely explored in various research publications. Studies revealed the relevance of machine learning techniques along with artificial intelligence for the recognition of human emotions such as sadness, anger, happiness, etc. using datasets like face image, person video, audio, questionnaire-response, etc. Human emotions come from psychological activities that may be affected by outside daily life routines. The proposed study reveals the configuration of anxiety and depression symptoms from the questionnaire-based dataset. In the present manuscript, we have used the standard DASS-21 questionnaire for the identification of anxiety and depression by applying machine learning algorithms on the user responses. We have analyzed and presented the comparative performance of five classification algorithms i.e., SVM, Decision Trees, Random Forest, Naïve Bayes and KNN on the aforementioned problem of identification of users under Depression and Anxiety.**

***Keywords*— DASS-21, Questionnaire method, Naïve Bayes, K-nearest neighbor, Decision tree, Random forest and Support vector machine.**

I. Introduction

Anxiety and depression are the negative psychological activities that occur in the human brain. Such activities may be triggered genetically or by physical surroundings. Emotional recognition models are developed to analyze emotions i.e. sad, angry, happy, excited, etc. Such a model identifies features of body signals contained in terms of facial expression, body movement, speech signal, blood pressure, etc. Various tools and devices are developed to record and process such body signals. The aim of the model is to accurately correlate body signals with psychological activities. Computer-based technology generally includes a machine learning/deep learning model for processing the features. The human body releases many chemicals and shows a change in non-verbal body language. Emotional change brings variation in the body's activities. Intelligent models are used to track the variation for the identification of the emotional type. Anxiety is a primary stage of disorder that is temporary in nature. This can occur due to small issues in the daily routine. Depression is the severe stage of psychological disorder that may show the long-term adverse effect on the person's mental and physical health. Various researches have been conducted for measuring emotional variation [1] through facial activities. Change in facial expression is correlated with psychological activities. Expression of anger, sadness, excitement, etc. is easily visible through the facial expression. But the study consumes more data size as it uses an image or video dataset. Such studies may not validate the identification results as the facial expression can be falsely generated or controlled. Some research works applied models that are able to process heart rate signals [2] and blood pressure [3] using specialized devices. Variation in heart rate and blood pressure gives authentic features for the identification of psychological activities. But such systems require expensive tools for data collection. Some researchers use patients' audio signals as a dataset [4] for measuring the correlation of variation in the frequency components with the psychological activities. Using audio/speech dataset, an another way to identify the stress, may not be found robust as such signals can be manipulated or faked. Thus questionnaire based identification seems to the an ideal choice. A questionnaire-based dataset such as the perceived stress scale (PSS) [5] is also tested in existing literature for the identification of psychological symptoms. The questionnaire method provides score points depending on the severity level of the symptoms that have been reflected through the responses.

II. Related Work

Sonhi et al. [15] explored a study in which the effect of stress activity has been relatively seen with the person's vocal intensity. In this study, features like pitch, frequency, energy, amplitude etc are extracted out in the form of frames in order to carry of measure of stress level. The study was also conducted to fetch long-term feature which includes speed of variations, range of signals etc. Sandulescu et al. [16] had investigated stress exploration using support vector machine that act a classifier. This classifier has used low-level feature components to obtain stress recognition. These low-level features contains medium range frequency frames and low level pitch voice. Hyewon et al. [17] uses LSTM (long short term memory) along with RNN (recurrent neural network) to identify the stress symptoms from speech signals. Contextual information has been stored in human speech that contains the relation between the subsequent speech frames. The contextual information has been extracted using coefficients of filter bank components. The coefficients are contained with spectral information of speech signal. The experiment was conducted in the relax environment in which the person has been asked few standard questions from the script. An audio recorder was placed to record the responses from person in the form of speech signal. Automatic analysis of depression has been observed using facial variation of subject. This method is very common and found to be robust to secure extraction and classification of level of stress intensity. Wen et al. [18] has proposed a methodology for the recognition of stress using the sequence of facial image. The SVR algorithm has been applied over LPQ-TOP facial features for the detection of depression from an individual. CNN (convolutional neural network) has been employed in various earlier techniques to carry of the extraction of features related to the depression and stress. The deep neural network was found to be very effective of large set of complex data. Such data is easily processed and classified using neural network. Al Jazaery et al. [19] had implemented feed forward deep neural network for the translation of depression and stress from video based features. The video based features are also tested with recurrent neural network in which the temporal information has been explored from the video frames.

The proposed model applied a questionnaire dataset in which standard question related to daily routine has been asked to the peoples. A person needs to reply to the questions in form of textual data. The standard questions fall under the DASS-42 dataset. The questionnaire-based dataset is intended to trigger psychological activities in the human mind. The responses of each user contain theoretical features of psychological activities. Working with the questionnaire dataset is less expensive in terms of memory consumption and time complexity. The proposed model applies features processing tasks with the help of machine learning algorithms. The model tested five different machine learning algorithms such as naïve Bayes, SVM, decision tree, random forest, and K-nearest neighbor. These algorithms are used for the classification of features. The algorithm finds a correlation between the score points decided by the DASS-42 dataset. All the correlated score points for the anxiety symptoms are mapped in one class. And all the correlated score points for the depression symptoms are mapped in different classes. Correlations of score points are measured in various ways inside the algorithms.

The chapter scheme of the proposed paper is as follows:- Section 3 contains the proposed methodology in which SVM, decision tree, random forest, KNN, and naïve Bayes algorithms are discussed. Section 4 is the result section that shows the effectiveness of the algorithm. Conclusion is presented in section 5 followed by references.

III. PROPOSED METHODOLOGY

The proposed methodology uses a questionnaire-based dataset for the identification of depression and anxiety symptoms. These symptoms are classified using various classifies.

A. Dataset

The proposed methodology uses DASS-42 [6] questionnaire-based dataset collected between 2017 and 2019 in the online and offline surveys. The dataset contains 39,777 sets of responses against questionnaires from each individual that are collected in the form of textual representation. DASS-42 dataset is a collection of 42 sets of standard questions. Each response from the individual against the questionnaire set has been mapped with the standard scaling ranging between 1 and 4.

The specification of scaling has been given below:-
Scale 1:- No anxiety/depression discovered
Scale 2:- Anxiety and depression found in some degree
Scale 3:- Moderate level of anxiety
Scale 4:- Severe level of anxiety and depression

In this dataset, all the questions are intended to find depression and anxiety symptoms. The individual responses are collected in scaling discussed above. The scale values of each response are added against the questionnaire. After the addition, the final score has been labeled with respect to the severity of the depression and anxiety. Figure 1 contains the score calculation against the DASS-42 questionnaire dataset for the identification of severity level of anxiety.

Q2	Q4	Q7	Q9	Q15	Q19	Q20	Q23	Q25	Q28	Score	Anxiety
2	2	2	0	1	3	3	0	2	1	16	Severe
2	1	3	3	3	2	2	3	0	2	21	Extreme Severe
1	0	0	0	2	0	6	2	0	3	26	Moderate
0	0	1	0	2	0	0	3	0	0	6	Normal
0	1	2	0	0	0	3	2	0	1	9	Mild

Table 1:- Sample of score calculation for anxiety

Figure 1 shows the sample of responses received against questions in the dataset. The scores are added respectively for the identification of severity level of anxiety. Figure 2 contains the score calculation against the DASS-42 questionnaire dataset for the identification of depression.

Q2	Q4	Q7	Q9	Q15	Q19	Q20	Q23	Q35	Q48	Score	Depression
3	3	0	1	1	2	1	3	1	3	17	Severe
3	2	2	2	3	3	3	2	3	2	25	Extreme Severe
1	0	0	1	2	3	2	3	1	2	15	Mild
0	0	1	0	2	3	2	3	2	3	16	Moderate
0	1	2	0	0	1	3	3	0	2	12	Normal

Table 2:- Sample of score calculation for depression

Figure 2 shows the sample of responses received against questions in the dataset. The scores are added respectively for the identification of severity level of depression. Figure 3 shows the flow chart of the proposed methodology.

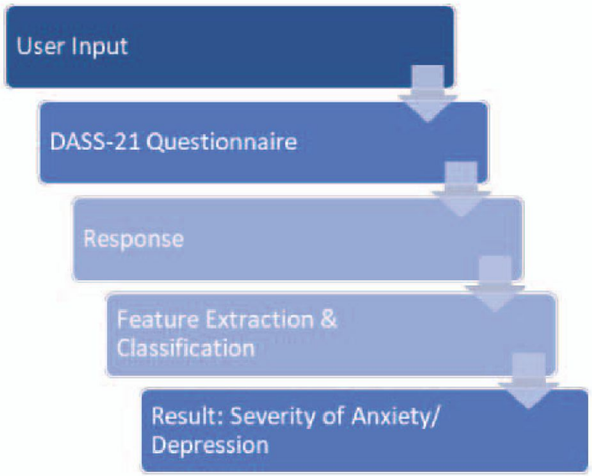

Figure 1:- Flow chart of the proposed methodology

Figure 1 shows the working of the proposed methodology. The proposed model describes that the users having anxiety and depression disorder have participated in the study. Responses of each user have been collected against the questionnaire dataset. Then, feature extraction has been done from the responses. Features of each user response are in the form of score values that are mapped with responses as standardized by the DASS-42 dataset. These scores are then classified into depression and anxiety symptoms. The proposed model applied machine learning algorithms including SVM, naïve Bayes, decision tree, random forest, and KNN. These algorithms are individually applied over the features of responses in order to obtain the classification. The scaling parameter of the DASS-42 questionnaire dataset is much similar to the scaling factor of the PSS (perceived stress scale) method in which responses against the questionnaire has been mapped with score ranging from 0 to 4. These score values are used to determine the severity level of disorder symptoms. The questionnaire approach is totally based on the user response.

B) Classification algorithms

The DASS-42 score factors contain data classified into four scaling points' ranges from 1 to 4. The stratified data in these ranges are further sent into the classification algorithm for the detection of anxiety and depression. The proposed model uses five classification algorithms i.e. Naïve Bayes, K-nearest neighbor, Decision tree, random forest, and Support vector machine. Descriptions of these classifiers are shown in this section below.

a) Naïve Bayes

This algorithm is based on a probabilistic approach for the identification of classes. It uses Bayes theorem to find the probability of belongingness of a feature space into a class that could be anxiety and depression. The classifier takes input from DASS-21 scaling factors and analyses it using the Bayes theorem.
The equation of Bayes theorem is defined as:-

$$P(Y/Z) = \frac{P(Z|Y)P(Y)}{P(Z)}$$

Here,

- *P(Y|Z) is the probability of X such that event B is already true.*
- *P(Y) is the prior probability of class.*
- *Y and Z are the two events.*
- $P(Z|Y)$ *is the likelihood which is the probability of predictor given class.*

The naive Bayes classifier compares the probabilities of belongingness of various feature spaces in a class. The feature that has maximum probability is classified into a specific class. The dataset has been divided into 70:30 for training and testing respectively.

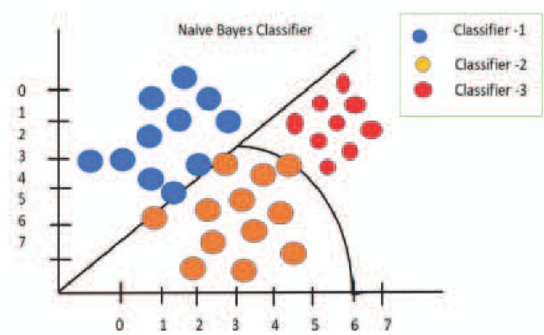

Figure 2:- Representation of Naïve bayes classifier

Figure 2 shows the representation of naïve Bayes classifier which is based on probabilistic classification of data samples.

b) K-nearest neighbor (KNN)

This algorithm works on the basis of Euclidean distance in which the neighboring features are found out. Features of anxiety must contain the nearest distance and the same applies to the features of depression. The distance algorithm has been applied in DASS records and the textural data has been classified into depression and anxiety. The steps of the algorithms are described below.

Step 1:- Input DASS-42 records
Step 2:-The similarity between two selected textural data has been measured by using the following equations:

$$Image(A) = \sum_{m=0}^{m} \sum_{n=0}^{n} Z_x {Z'}_y$$

$$Distance_1(Img_x Img_y) = \sum_{m=1}^{N} |Z_m - {Z'}_m|$$

$$Distance_2(Img_x Img_y) = \sqrt{\sum_{m=1}^{N} |Z_m - {Z'}_m|}$$

$$Distance_2(Img_x Img_y) = \sqrt{\sum_{m=1}^{N} |Z_m - Z'_m|}$$

$$Distance_{cos}(Img_x Img_y) = 1 - \frac{\overrightarrow{Z_m} \cdot \overrightarrow{Z'_m}}{|Z_m| \cdot |Z'_m|}$$

Step 3:- Return
Step 4:- Accuracy measure

$$I(A) = \sum_{m=0}^{m} \sum_{n=0}^{n} Img_x Img_y$$

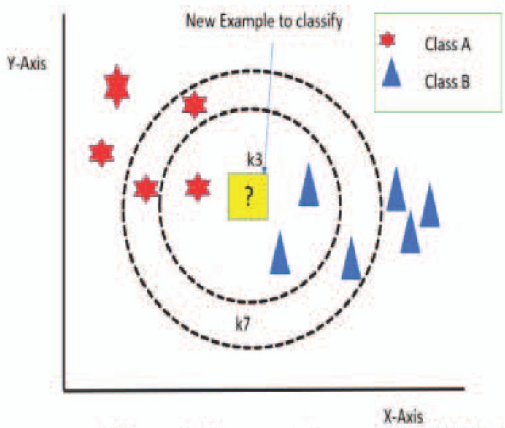

Figure 3:- Representation sample of KNN

Figure 3 shows the representation of the KNN algorithm in which two classes are shown. The first class has been shown with the representation of start (red color) and the second class is represented by a triangle (blue color). The objects of the first class are nearer to each other and the same has been seen in other classes also.

c) Decision Tree

The concept of decision is used to take decisions for the symptoms either they are depression or anxiety. The symptoms are collected in the form of responses against the DASS-21 questionnaire. The algorithm of the decision tree has been described below.

Suppose $d_1, d_2, d_3 \dots d_n$ are the data feature vectors of DASS-21 scale.

Step 1:- Entropy is the impurity present in the dataset. The aim is to find the information of all the attributes of the dataset.

$$Entropy = \sum_{i=1}^{3} -P(x)_i \; log_2 P(x)_i'$$

P(x) is the probability of an attribute of data being belong of stress calls. The iteration of variable "i" ranges from 1 to 3 for stress depression and anxiety.

Step 2:- Then, information of each attribute has been calculated.

$$Information(I(x)) = E(x) - W_A \times E(x)$$

Here, W_A is the weighted average of the sub-attributes of an attribute

Step 3:- Now, Information gain will be calculated which shows the information of the entire single attribute relative of the entire dataset.

$$Information\ gain = E(S) - Img(x)$$

The E(S) is the entropy of a whole single attribute relative to dataset S. The Img(x) is the information that attribute.

d) Random Forest

The concept of the random forest has been derived from the decision tree. A decision tree is used to make decisions for a given dataset. The random forest algorithm is applied when the dataset becomes complex and so multiple decisions are required to process. There, individual decision trees are generated and combined to form the random forest. The leaf node always contains the decision related to anxiety and depression.

Suppose $A_1, A_2, A_3 \dots A_n$ are the data feature vectors of DASS-21 scaling points.

Step 1:-

$$A = \frac{1}{N} \sum_{i=1}^{N} A_i$$

Step 2:- Subtract the mean:-

$$\lambda_n = A_i - \hat{a}$$

Step 3:- For feature matrix Y = $[\lambda_1, \lambda_2, \dots\dots \lambda_n]$

$$A = \frac{1}{N} \sum_{N=1}^{N} \lambda_n \lambda_n^{\ T} = \frac{1}{N} YY^T$$

Step 4:- Compute eigenvalues

A: $\beta_1 > \beta_2 > \cdots \beta_n$

$\beta_1 > \beta_2 > \cdots \beta_n$ are orthogonal values of input data matrix. $A - A'$ is the linear combination of eigenvectors.

$$A - A' = b_1 \Pi_1 + b_2 \Pi_2 + \cdots b_n \beta_n = \sum_{i=1}^{N} b_n \beta_n$$

$$b_i = \frac{(A - A').\beta_n}{\beta_n^2}$$

Step 5:- Information combined:-

$$Info = \sum_{k=1}^{H} b_n \beta_n$$

Here, Info is the highest eigenvalue.

e) Support Vector Machine

Support vector machine is a machine learning model that is used to draw a hyper-plane that separates two kinds of classes. The proposed model applies SVM to classify the features of depression and anxiety through the hyper-plane slope. The proposed model applied default kernel settings depending on the dataset for the classification of response-based features.

IV. Result discussion

The result has been conducted after applying the machine learning algorithms over the responses against the dataset. Table 1 contains the confusion matrix obtained by all the respective machine learning algorithms for anxiety and depression analysis. The confusion matrix shows the recognition accuracy of any anxiety and depression symptoms for which the user responses are processed. The confusion matrix shows the evidence of true positive rate, true negative rate, false positive rate, and the false negative rate.

Table 1. Confusion matrices for anxiety and depression

Method	Anxiety					Depression				
Naïve Bayes	19	0	0	0	0	28	0	0	0	0
	0	35	22	0	0	6	26	22	0	0
	0	15	2	0	0	0	7	24	3	0
	0	5	0	16	0	0	5	0	14	0
	0	0	0	8	4	0	0	0	8	7
Decision Tress	34	0	0	0	0	28	0	0	0	0
	9	27	9	0	0	6	26	9	0	0
	0	28	38	4	0	0	7	24	3	0
	0	5	0	1	0	0	5	0	14	0
	0	0	8	16	4	0	0	7	1	4
Random forest	33	0	0	5	0	18	0	0	5	0
	9	29	15	0	0	9	23	0	0	0
	0	0	18	7	5	0	7	14	6	5
	0	5	4	12	0	0	5	4	7	3
	0	0	3	1	4	0	0	0	0	9
KNN	25	0	0	5	0	18	0	0	4	2
	9	23	15	0	0	9	29	15	0	0
	0	18	13	7	5	0	18	23	3	5
	5	5	4	19	0	5	1	7	17	0
	0	0	3	1	2	0	0	3	2	2
SVM	22	0	0	4	8					
	9	27	1	0	0	26	0	0	4	8
	0	18	23	7	5	9	22	6	0	0
	5	0	0	29	0	0	18	16	2	5
	0	0	0	0	2	8	2	0	21	0
						0	0	2	0	6

The above table is showing confusion matrices for anxiety, depression, and stress. These confusion matrices are generated by all the five classifiers used in the proposed scheme. Table 2 shows the measurement of various parameters for all the classifiers used in the study. Based on the given parameters, the overall efficiency of each model has been described.

Table 2. Measure of accuracy, error rate, precision, recall and F1 score for anxiety.

Classifier	Accuracy	Error Rate	Precision	Recall	F1
Naïve Bayes	0.738	0.270	0.523	0.658	0.628
Decision tree	0.799	0.23	0.79	0.57	0.53
Random forest	0.78	0.26	0.73	0.69	0.68
KNN	0.831	0.23	0.75	0.84	0.81
SVM	0.913	0.25	0.62	0.83	0.82

Table 3. Measure of accuracy, error rate, precision, recall and F1 score for depression

Classifier	Accuracy	Error Rate	Precision	Recall	F1
Naïve Bayes	0.851	0.28	0.574	0.798	0.758
Decision tree	0.78	0.24	0.77	0.59	0.52
Random forest	0.76	0.22	0.73	070	0.66
KNN	0.856	0.25	0.853	0.896	0.821
SVM	0.912	0.26	0.63	0.84	0.82

The above table shows the various quality parameters containing accuracy, error rate, precision, recall, and F1 score for the respective anxiety (A), depression (D), and stress (S). These measures are individually calculated by all four types of classifiers.

Table 3:- Accuracy comparison of applied machine learning model

Classifier	Accuracy
Naïve Bayes	82%
Decision Tress	86%
Random Forest	87%
KNN	89%
SVM	91%

Table 3 shows the tabular comparison of average accuracies obtained by all the five classifiers for depression and anxiety.

V. Conlcusion

The proposed model illustrates the working of five machine learning classifiers to conduct an experiment for the detection of depression and anxiety. The model accomplishes an objective to recognize the relation of psychological states of people with their responses on DASS-21 questionnaire, accurately. The dataset used in the model is a set of questionnaires named DASS-42, which has been extensively used in the past for the identification of depression and anxiety, manually . The proposed system obtains the responses on questionnaires from various people through offline/online interrogation. The responses are processed and analysed using a standard scaling factor. Then, classification algorithms have been applied to classify the data into depression and anxiety. From the result section, it is concluded that the SVM based model is most efficient for the detection of depression and anxiety.

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